Pass-Through, Welfare, and Incidence under Product Returns in Perfect Competition

Draft notes

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1 Proposal

About one in five products sold online is returned. Pass-through, tax incidence, and the efficiency cost of taxation are well understood when purchase settles who keeps the good. In that classical setting, the demand and supply curves are sufficient for consumer surplus, producer surplus, and deadweight loss. With returns, allocation is decided in two stages. Buyers first choose whether to purchase, then whether to keep or return after receiving the item. Never buying and buying then returning leave the same final allocation—the consumer does not keep the good—yet their welfare costs differ. A returned purchase incurs shipping, hassle, and handling that a non-purchase avoids, so demand and supply alone omit an economically relevant margin.

What objects then replace the classical demand and supply curves, and how misleading is an analysis that ignores returns? We study these questions under perfect competition with product returns and seller handling costs. We characterize pass-through, incidence, and welfare losses with returns, compare them to the no-return benchmark, and identify where omitting the return margin changes conclusions.

Consumer surplus depends only on two aggregates: gross demand and the return rate. Heterogeneity in tastes, match values, and return costs enters welfare solely through these objects. Producer surplus retains the usual geometry—area to the left of the supply curve—but evaluated at net revenue rather than at list price. Net revenue is list price minus expected refunds and handling.

Because consumers pay the list price while firms respond to net revenue, pass-through from cost shocks into list prices is governed by how net revenue moves with the list price. Tax neutrality fails for the same reason. The keep-or-return decision depends on the refund price, which differs according to which party remits the tax. An analyst who ignores returns and applies the classical formulas therefore misstates incidence and deadweight loss whenever the return margin is active. List prices do not equal firm net revenue, and consumer surplus tracks kept units rather than gross sales.

2 Setting

A homogeneous product is sold at list price p. Consumers indexed by i differ in return hassle κ_i, the disutility from returning the product, and in match distribution G_i over post-purchase match quality u. After purchase, type i draws $u \sim G_i$ and keeps the item whenever $u - p \geq -\kappa_i$, earning surplus $u - p$, and returns it otherwise, earning $-\kappa_i$.

Buying at price p yields expected surplus

$$s_i(p) = \mathbb{E}_{u \sim G_i}\big[\max(u - p,\, -\kappa_i)\big].$$

On the purchase margin, all consumer-side heterogeneity is summarized by willingness to pay v_i, defined by $s_i(v_i) = 0$, so type i buys if and only if $p \leq v_i$. Across types, willingness to pay has distribution $F(v) = \Pr(v_i \leq v)$, so gross demand is $D(p) = 1 - F(p)$. Of the gross units sold at price p, a share $r(p)$ are returned.

We do not impose a sign on how $r(p)$ varies with p. A higher price screens out low-value buyers and so tends to lower the average return rate among purchasers, but it also raises each buyer's keep threshold and therefore raises individual return probabilities. These selection and behavioral effects pull in opposite directions.

The product market is perfectly competitive, producer j takes p as given, chooses output at cost $C_j(q_j)$, and pays handling cost $h > 0$ on each returned unit.

3 Consumer side

Write $s(v, p)$ for expected surplus at list price p among types with willingness to pay v, so $s(v, v) = 0$. Consumer surplus integrates surplus over all types who buy,

$$CS(p) = \int_p^\infty s(v, p)\, dF(v). \tag{1}$$

Differentiating in p gives an extensive-margin term from types entering or leaving the purchase margin and an intensive-margin term from lower surplus among inframarginal buyers,

$$CS'(p) = -s(p, p)\, f(p) + \int_p^\infty s_p(v, p)\, dF(v),$$

where $s_p(v, p) \equiv \partial s(v, p)/\partial p$ and f is the density of v_i when F is differentiable. On the extensive margin, $s(p, p) = 0$, so entry and exit contribute nothing at first order. On the intensive margin, $s_p(v, p)$ equals minus the keep probability, because a higher price harms the consumer only when she keeps the purchase,

$$s_p(v, p) = -\Pr(u \geq p - \kappa_i).$$

Aggregating over buyers who purchase then gives

$$CS'(p) = \int_p^\infty s_p(v, p)\, dF(v) = -D(p)\big(1 - r(p)\big),$$

by the definitions of $D(p)$ and $r(p)$. A small price change therefore satisfies

$$dCS = -D(p)\big(1 - r(p)\big)\, dp. \tag{2}$$

The extensive- and intensive-margin decomposition makes the local formula transparent. Consumer surplus changes depend only on the aggregates $D(p)$ and $r(p)$, both at each price and integrated along any path,

$$\Delta CS = -\int_{p_0}^{p_1} D(p)\big(1 - r(p)\big)\, dp.$$

Micro heterogeneity in price responses can be rich. Price p enters each $s_i(p)$ nonlinearly and can shift participation, return decisions, and inframarginal surplus differently across types. Yet once

(D, r) are given, this microstructure is irrelevant for dCS and ΔCS. Two economies with the same paths of $D(p)$ and $r(p)$ deliver the same consumer surplus loss, whether high-v_i types adjust return behavior more steeply than low-v_i types or not. Only kept units enter at first order, through $1 - r(p)$, not gross purchases $D(p)$ alone.

Let $\bar{p}$ denote the highest willingness to pay in the population, so $D(\bar{p}) = 0$ and $CS(\bar{p}) = 0$. Integrating the local formula then rewrites consumer surplus as area under kept volume,

$$CS(p) = \int_p^{\bar{p}} D(z)\big(1 - r(z)\big)\, dz. \tag{3}$$

4 Supply side

Returns make list price an imperfect summary of producer payoffs. A gross sale brings revenue p if the consumer keeps the item and costs refund p plus handling h if the item is returned. Among gross sales at list price p, share $r(p)$ are returned, so expected revenue per gross unit—the *effective marginal revenue*—is

$$\mu(p, h) \equiv p - r(p)(p + h). \tag{4}$$

Under perfect competition, firm j takes p as given and therefore faces a fixed number μ in its supply decision. At equilibrium, $\mu = \mu(p, h)$.

Given μ, firm j chooses output to solve

$$\max_{q_j \geq 0} \ \mu\, q_j - C_j(q_j), \qquad \text{FOC: } C_j'(q_j) = \mu \ \ (\text{if } q_j > 0),$$

with $C_j(0) = 0$. Write $q_j(\mu)$ for firm j's supply, set to zero when μ is below its minimum marginal cost.

Summing across firms defines market supply and industry profit,

$$S(\mu) \equiv \sum_j q_j(\mu), \qquad PS(\mu) \equiv \sum_j \big[\mu\, q_j(\mu) - C_j\big(q_j(\mu)\big)\big]. \tag{5}$$

Which firms are active at each μ determines both the market quantity and total profit.

Differentiating $PS(\mu)$ with respect to μ gives, for each firm,

$$\frac{d}{d\mu}\Big[\mu\, q_j - C_j(q_j)\Big] = q_j + q_j'(\mu)\,\big[\mu - C_j'(q_j)\big].$$

Active firms satisfy $C_j'(q_j) = \mu$ at their chosen output, so the adjustment term $q_j'(\mu)\,[\mu - C_j'(q_j)]$ is zero for each firm—equivalently, the envelope theorem applied to each firm's profit problem gives $d\pi_j/d\mu = q_j(\mu)$. Summing over firms,

$$PS'(\mu) = \sum_j q_j(\mu) = S(\mu).$$

With no production at $\mu = 0$, it follows that industry profit equals the area to the left of the market supply curve,

$$PS(\mu) = \int_0^{\mu} S(z)\, dz. \tag{6}$$

At equilibrium $\mu = \mu(p, h)$, $PS = \int_0^{\mu(p,h)} S(z)\, dz$.

5 Failure of tax neutrality

Weyl and Fabinger (2013) show that without returns, buyer and seller remittance deliver the same equilibrium. With returns, we ask whether that equivalence survives by writing the two clearing conditions at a common checkout price p—the total payment at the register—and a per-unit tax τ. Fix p and τ. Seller remittance clears at

$$D(p) = S\big(p - \tau - r(p)(p + h)\big), \tag{7}$$

because both the return rate and per-return cost $p + h$ are evaluated at checkout price p. Buyer remittance clears at

$$D(p) = S\big((p - \tau) - r(p - \tau)(p - \tau + h)\big), \tag{8}$$

because the seller posts list price $p - \tau$, so returns and refund cost are evaluated there. The two conditions define the same equilibrium only if the arguments of $S(\cdot)$ coincide. Canceling $p-\tau$ yields

$$r(p)(p + h) = r(p - \tau)(p - \tau + h). \tag{9}$$

Result 1 (Failure of tax neutrality with returns)**.** *Seller and buyer taxes define the same equilibrium if and only if* $r(p)(p + h) = r(p - \tau)(p - \tau + h)$.[1]

Tax neutrality breaks down with product returns because the refunded amount—and therefore the keep-or-return decision—depends on which side of the market remits the tax. Under seller remittance, the checkout price (which includes the tax) is refunded, so consumers decide whether to return based on that tax-inclusive price. Under buyer remittance, the seller posts a pre-tax list price, only that list price is refunded, and the return decision is based on this lower, tax-exclusive price. Even though the total checkout price and the tax are identical in both regimes, the return rate and the seller's refund liability per return are evaluated at different prices (p versus $p - \tau$).

The direction in which the return rate changes is ambiguous—a higher price can make each individual more likely to return but can also select consumers with stronger purchase intentions who are less likely to return—so $r(p)$ may be above or below $r(p - \tau)$. Regardless of the direction, the seller's effective marginal revenue generally differs, so the two tax regimes produce distinct market-clearing conditions and are not equivalent. The fundamental cause is that the purchase decision is always based on the checkout price, while the return decision uses the tax-inclusive price under seller remittance and the tax-exclusive price under buyer remittance, creating a wedge that alters equilibrium.

6 Pass-through

Write $Q \equiv D(p)$. We first study pass-through for two cost shocks: a uniform increase in marginal production cost (denoted c) and an increase in the per-unit handling cost h; the same framework is then extended to a per-unit tax. A uniform production-cost increase means adding one dollar to each firm's marginal cost $C_j'(q_j)$ at every output level. Production cost and handling cost hit equilibrium on different margins. Production cost shifts market supply at a given μ without moving μ at fixed list price. Handling cost moves μ through (4) at fixed list price without shifting supply at a given μ.

[1] Neutrality requires the return wedge to be the same at checkout price p and at posted list price $p-\tau$. That holds for all taxes only if $r(p)(p+h)$ is constant in posted list price, which requires $r(p) = C/(p+h)$ for some constant C; without that functional form on r, the two clearing conditions need not define the same equilibrium.

Equilibrium satisfies

$$m(p,x) \equiv D(p) - S(\mu(p,x),x) = 0, \qquad x \in \{c,h\}. \tag{10}$$

Differentiating while keeping $m = 0$ defines pass-through $\rho_x \equiv dp/dx$ by

$$\rho_x = -\frac{m_x}{m_p}, \tag{11}$$

where $m_p \equiv \partial m/\partial p$ and $m_x \equiv \partial m/\partial x$. Since $m = D - S$,

$$m_p = D'(p) - S'(\mu)\,\mu_p, \qquad\qquad m_x = -S'(\mu)\,\mu_x - S_x, \tag{12}$$

with $\mu_p \equiv d\mu/dp$, $\mu_x \equiv \partial\mu/\partial x$ at fixed p, $S'(\mu) \equiv dS/d\mu$, and $S_x \equiv \partial S/\partial x$ at fixed μ. Hence

$$\rho_x = \frac{S'(\mu)\,\mu_x + S_x}{D'(p) - S'(\mu)\,\mu_p}. \tag{13}$$

At fixed list price p, a one-dollar parallel rise in production cost shifts supply at fixed μ, so $\mu_c = 0$ and $S_c = -S'(\mu)$. A one-dollar rise in handling cost moves μ by $r(p)$ per gross unit sold and does not shift supply at fixed μ, so $\mu_h = -r(p)$ and $S_h = 0$. Hence

$$\mu_c = 0, \qquad S_c = -S'(\mu), \qquad \mu_h = -r(p), \qquad S_h = 0. \tag{14}$$

Substituting into (13) yields

Result 2 (Pass-through of production and handling costs)**.**

$$\frac{\rho_h}{\rho_c} = r(p), \tag{15}$$

with pass-through into list price

$$\rho_c = \frac{-S'(\mu)}{D'(p) - S'(\mu)\,\mu_p}, \qquad \rho_h = r(p)\,\rho_c. \tag{16}$$

A one-dollar increase in production cost is equivalent to decreasing μ by one unit. Market clearing $D(p) = S(\mu)$ translates a given change in μ into list price through the same local comparative statics. Handling moves μ by only $r(p)$ per dollar relative to that unit shift, so $\rho_h/\rho_c = r(p)$. When $r \equiv 0$, handling does not move μ and ρ_c coincides with the textbook competitive pass-through formula.

Write p^s for checkout price under seller remittance and p^b for checkout price under buyer remittance; let $\rho^s_\tau \equiv dp^s/d\tau$ and $\rho^b_\tau \equiv dp^b/d\tau$. For a local change at $\tau = 0$, tax pass-through is (13) with $S_\tau = 0$: the denominator uses the same μ_p as above, and the only difference across remittance regimes is $\mu^s_\tau \equiv \partial\mu^s/\partial\tau$ versus $\mu^b_\tau \equiv \partial\mu^b/\partial\tau$ at fixed checkout price:

$$\rho^s_\tau = \frac{S'(\mu)\,\mu^s_\tau}{D'(p^s) - S'(\mu)\,\mu_p}, \qquad \rho^b_\tau = \frac{S'(\mu)\,\mu^b_\tau}{D'(p^b) - S'(\mu)\,\mu_p}. \tag{17}$$

At $\tau = 0$,

$$\mu^s_\tau = -1, \qquad\qquad \mu^b_\tau = r(p^b) + (p^b + h)\,r'(p^b) - 1. \tag{18}$$

Under seller remittance, $\mu^s_\tau = -1$ at fixed checkout price, so $\rho^s_\tau = \rho_c$. Under buyer remittance, μ^b_τ differs, so tax pass-through generally differs from ρ_c.

Elasticity representation and interpretation.
The level-form expression for production-cost pass-through in (16) is compact, but translating it into conventional elasticities reveals the precise economic mechanism. Define the demand elasticity with respect to the list price,

$$\varepsilon_D(p) \equiv \frac{pD'(p)}{D(p)} < 0,$$

and, following the standard convention, define the supply elasticity with respect to the list price p:

$$\varepsilon_S(p) \equiv \frac{dS(\mu(p))}{dp} \frac{p}{S(\mu(p))} = S'(\mu)\,\mu_p \,\frac{p}{S(\mu)} > 0.$$

This is the observed elasticity of market supply with respect to the price that clears the market. From $\mu = p - r(p)(p+h)$, the sensitivity of effective revenue to the list price is

$$\mu_p \equiv \frac{d\mu}{dp} = 1 - r(p) - r'(p)(p+h)$$

where $r'(p) \equiv dr(p)/dp$. Substituting these definitions into ρ_c in (16) and using $D(p) = S(\mu)$ at equilibrium gives

$$\rho_c = \frac{-S'(\mu)}{D'(p) - S'(\mu)\mu_p}. \tag{19}$$

Multiplying the numerator and denominator by $p/S(\mu)$:

$$\rho_c = \frac{-S'(\mu)\,p/S(\mu)}{\varepsilon_D - S'(\mu)\mu_p\,p/S(\mu)}. \tag{20}$$

Since $\varepsilon_S(p) = S'(\mu)\mu_p\,p/S(\mu)$, we have $S'(\mu)\,p/S(\mu) = \varepsilon_S(p)/\mu_p$. Therefore,

$$\boxed{\rho_c = \frac{-\varepsilon_S(p)/\mu_p}{\varepsilon_D - \varepsilon_S(p)}} \tag{21}$$

with μ_p as given above.

The elasticity expression (21) admits a sharp comparative interpretation. In the textbook competitive model without returns, supply is written directly as a function of the list price p, and a uniform cost increase shifts that curve upward. The standard formula is

$$\rho_{\text{std}} = \frac{-\varepsilon_S}{\varepsilon_D - \varepsilon_S}.$$

With returns, the supply curve is fundamentally a function of the seller's effective revenue μ, not p. Nevertheless, by defining the observed supply elasticity $\varepsilon_S(p)$ with respect to the list price p—which inherently accounts for the pass-through μ_p from p to μ—the denominator of the pass-through formula retains the textbook structure $\varepsilon_D - \varepsilon_S(p)$.

The novelty therefore resides entirely in the numerator, which becomes $-\varepsilon_S(p)/\mu_p$ rather than $-\varepsilon_S(p)$. This extra factor of $1/\mu_p$ arises because the cost shock shifts the supply curve in the μ-space, not the p-space. Since $d\mu = \mu_p\,dp$, a unit leftward shift of the effective supply curve translates into a shift of size $1/\mu_p$ in the observed p–Q plane.

Consequently, the pass-through of production costs is amplified whenever $\mu_p < 1$—that is, whenever a higher list price raises the seller's net revenue by less than one dollar due to higher expected refunds. In the special case of no returns, $\mu_p = 1$ and $\varepsilon_S(p)$ coincides with the standard supply elasticity, collapsing (21) exactly to the textbook rule. This decomposition reveals that the economic logic of competitive pass-through is unchanged; returns merely rescale the effective supply response through the lens of the net-revenue pass-through μ_p.

7 Incidence

With the pass-through results in hand, we can now compute the welfare effect of each cost shifter and the division of the loss between consumers and producers. Throughout, $Q \equiv D(p)$ and $r \equiv r(p)$ denote the equilibrium gross quantity and the return rate. From the consumer-side analysis, a small increase in any shifter x changes consumer surplus by

$$\frac{dCS}{dx} = -Q\,(1-r)\,\rho_x, \tag{22}$$

where $\rho_x \equiv dp/dx$ is the pass-through of the shock into the list price.

Producer surplus is less immediate because a shock may shift the supply schedule directly as well as indirectly through the effective marginal revenue μ. Let $S(\mu, x)$ be the market supply curve, where the second argument allows for a direct effect of x on firms' costs, and write producer surplus as $PS(\mu, x) = \int_0^{\mu} S(z, x)\,dz$. Differentiating with respect to x,

$$\frac{dPS}{dx} = S(\mu, x)\,\frac{d\mu}{dx} + \int_0^{\mu} \frac{\partial S(z, x)}{\partial x}\,dz.$$

Using the equilibrium condition $S(\mu, x) = Q$ and denoting $\Sigma_x \equiv \int_0^{\mu} (\partial S(z, x)/\partial x)\,dz$, we obtain

$$\frac{dPS}{dx} = Q\,\frac{d\mu}{dx} + \Sigma_x = Q\big(\mu_p\,\rho_x + \mu_x\big) + \Sigma_x, \tag{23}$$

with $\mu_p \equiv \partial\mu/\partial p$ and $\mu_x \equiv \partial\mu/\partial x$ evaluated at fixed list price.

The term Σ_x captures the direct shift of the supply curve holding μ constant. A uniform, parallel increase in every firm's marginal production cost by one dollar ($x = c$) moves the supply curve inward according to $\partial S(\mu, c)/\partial c = -S'(\mu)$, so $\Sigma_c = \int_0^{\mu} (-S'(z))\,dz = -S(\mu) = -Q$. For a handling-cost shock h or a per-unit tax (whether seller- or buyer-remitted), the supply schedule at a given μ is unchanged, hence $\Sigma_h = \Sigma_{\tau_s} = \Sigma_{\tau_b} = 0$. The partial derivatives μ_x are obtained directly from the definition of effective marginal revenue: $\mu_c = 0$, $\mu_h = -r$, $\mu_{\tau_s} = -1$, and, for the buyer-remitted tax evaluated at $\tau = 0$, $\mu_{\tau_b} = r + (p+h)r' - 1$ as derived in Section 5.

Substituting these values together with the pass-through expressions $\rho_h = r\rho_c$ and $\rho_\tau^s = \rho_c$ into (23) yields the producer-surplus responses:

$$\frac{dPS}{dc} = Q\big(\mu_p\,\rho_c - 1\big), \tag{24}$$

$$\frac{dPS}{dh} = Q\big(\mu_p\,\rho_h - r\big), \tag{25}$$

$$\frac{dPS}{d\tau_s} = Q\big(\mu_p\,\rho_c - 1\big), \tag{26}$$

$$\frac{dPS}{d\tau_b} = Q\big(\mu_p\,\rho_\tau^b + \mu_{\tau_b}\big). \tag{27}$$

The incidence ratio $I(x) \equiv |dCS/dx| \,/\, |dPS/dx|$ measures how the total loss is divided. For production cost, handling cost, and the seller-remitted tax, the ratio simplifies to the same expression:

$$I(c) = I(h) = I(\tau_s) = \frac{(1-r)\,\rho_c}{1 - \mu_p\,\rho_c}. \tag{28}$$

For the buyer-remitted tax, the ratio is

$$I(\tau_b) = \frac{(1-r)\,\rho_\tau^b}{-\big(\mu_p\,\rho_\tau^b + \mu_{\tau_b}\big)}, \qquad \mu_{\tau_b} = r + (p+h)r' - 1. \tag{29}$$

Equation (28) generalises the standard competitive-market formula $I = \rho/(1-\rho)$ (Weyl and Fabinger, 2013). Consumer loss is scaled by the keep probability $1-r$, and the producer-side denominator involves the effective-revenue pass-through $\mu_p \rho_c$ rather than the list-price pass-through alone. Production cost, handling cost, and a seller-remitted tax all generate the same incidence split because none of them changes the price at which consumers decide whether to return the product; they shift equilibrium exclusively through the effective marginal revenue μ. The buyer-remitted tax, in contrast, alters the refund price to the pre-tax list price, so the return rate and the per-return refund liability differ. This breaks the symmetry, giving an incidence ratio that departs from (28)—a welfare counterpart to the tax-neutrality failure established earlier. When returns are absent ($r = 0$, $\mu_p = 1$, $\rho_\tau^b = \rho_c$), all four ratios collapse to $\rho_c/(1-\rho_c)$, restoring the familiar equivalence.